INDIAN DIABETES PREDICTION

Exploratory Data Analysis & Machine Learning Classification

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768
Patient Records

34.9%
Diabetic Rate

9
Attributes

Pima Indians Diabetes Dataset (PIDD) • National Institute of Diabetes and Digestive and Kidney Diseases

AGENDA

01 Introduction

Understanding Diabetes — Types & Overview

02 Problem Statement

Key research questions to answer

03 Dataset

PIDD — 768 records, 9 attributes

04 EDA

Univariate, Bivariate & Multivariate Analysis

05 Hypothesis Testing & ANOVA

Statistical significance of features

06 ML Classification

SVM, Logistic Regression, Random Forest

07 Conclusion

Key findings & recommendations

INTRODUCTION

Diabetes is a chronic metabolic disorder where the body has high sugar levels for prolonged periods. It affects how the body turns food into energy. The lack of insulin is a primary cause.

Type I Diabetes

5–10%

Caused by autoimmune reaction — the body attacks the pancreas and stops insulin production. Requires insulin therapy.

Affects 5–10% of all diabetic patients globally

Type II Diabetes

90–95%

Body doesn't use insulin well and can't keep blood sugar at normal levels. Linked to lifestyle and obesity.

Affects 90–95% of all diabetic patients globally

DATASET

Pima Indians Diabetes Dataset • 768 Instances • 9 Attributes

Pregnancies int64

Number of times pregnant

Glucose int64

Oral Glucose Tolerance Test result

BloodPressure int64

Diastolic blood pressure (mm Hg)

SkinThickness int64

Triceps skin fold thickness (mm)

Insulin int64

Insulin levels in the blood

BMI float64

Body Mass Index (kg/m^2)

DiabetesPedigreeFunction float64

Family history likelihood score

Age int64

Age of the patient in years

Outcome int64

Target: 1 = Diabetic, 0 = Non-diabetic

EDA — STATISTICS SUMMARY

Variable	Mean	Std Dev	Min	Max	Skew
Pregnancies	3.85	3.37	0	17	0.90
Glucose	120.89	31.97	0	199	0.17
BloodPressure	69.11	19.36	0	122	-1.84
SkinThickness	20.54	15.95	0	99	0.11
Insulin	79.80	115.24	0	846	2.27
BMI	31.99	7.88	0	67.1	-0.43
DPF	0.47	0.33	0.08	2.42	1.92
Age	33.24	11.76	21	81	1.13

Zero Values Found

Glucose, BP, Skin, Insulin, BMI contain invalid 0s — replaced with column mean

Key Finding

Insulin has highest skew (2.27) & largest outlier; Glucose nearest to normal distribution

Target Distribution

65.1% Non-Diabetic vs 34.9% Diabetic — dataset is imbalanced

EDA — KEY FINDINGS FROM VISUALIZATIONS

Glucose — HIGHEST

Massive separation between Diabetic & Non-Diabetic distributions

Diabetic mean: 141 vs Non-diabetic: 110. Strongest predictor of diabetes onset.

BMI — HIGH

Diabetic patients have consistently higher BMI

Diabetic mean BMI: 35.1 vs Non-diabetic: 30.3. Obese range strongly associated with diabetes.

Age — HIGH

Diabetic patients are older on average

Diabetic mean age: 37.1 vs Non-diabetic: 31.2. Risk increases significantly after age 37.

Pregnancies — MEDIUM

Diabetic women had more pregnancies than non-diabetics

Correlated with Age (r=0.54). Higher pregnancy count is a secondary risk indicator.

Insulin — MEDIUM

Higher insulin levels in diabetics; large spread

Most skewed feature (2.27). Very high variance — extreme outliers present in both groups.

Blood Pressure — LOW

Small difference; both groups near-normal distribution

Near-normal skew (-1.84 after imputation). Negative feature importance in some models.

CORRELATION ANALYSIS

	PGN	GLU	BP	ST	INS	BMI	DPF	AGE	OUT
PGN		0.13	0.21	0.01	-0.02	0.02	-0.03	0.54	0.22
GLU	0.13		0.22	0.16	0.40	0.23	0.14	0.27	0.49
BP	0.21	0.22		0.13	0.01	0.28	0.00	0.33	0.16
ST	0.01	0.16	0.13		0.24	0.54	0.15	0.03	0.18
INS	-0.02	0.40	0.01	0.24		0.19	0.16	0.04	0.18
BMI	0.02	0.23	0.28	0.54	0.19		0.15	0.03	0.31
DPF	-0.03	0.14	0.00	0.15	0.16	0.15		0.03	0.17
AGE	0.54	0.27	0.33	0.03	0.04	0.03	0.03		0.24
OUT	0.22	0.49	0.16	0.18	0.18	0.31	0.17	0.24	

Key Correlations

Glucose → Outcome
r = 0.49 Strongest predictor

Pregnancies ↔ Age
r = 0.54 Highest inter-feature correlation

BMI ↔ SkinThickness
r = 0.54 Fat reserve relationship

BMI → Outcome
r = 0.31 Obesity risk factor

Age → Outcome
r = 0.24 Age-related risk

No multicollinearity detected (all r < 0.70)

HYPOTHESIS TESTING & ANOVA

Test 1: Normality of Glucose

H_0: Glucose is normally distributed
H_1: Glucose is NOT normally distributed

Statistic: 35.373 p-value: 0.000

REJECTED

Test 2: Glucose ↔ Outcome Correlation

H_0: Glucose & Outcome are uncorrelated
H_1: There is significant correlation

r = 0.493 p-value: 0.000

REJECTED

ANOVA — Analysis of Variance

793.31	0.000	1.940	H_0 Rejected
F-Test Statistic	p-Value	Critical F-Value	Conclusion

KEY QUESTIONS & ANSWERS

Q1: Chances of diabetes for age > 37?

51.49%

Over half of patients aged 37+ in the dataset are diabetic — age is a strong risk factor

Q2: Chances of diabetes if glucose > 140?

68.75%

Nearly 7 in 10 patients with glucose above 140 are diabetic — glucose is the #1 predictor

Q3: Best classification model accuracy?

SVM: 83%

Support Vector Machine outperformed all classifiers — robust to the outliers in this dataset

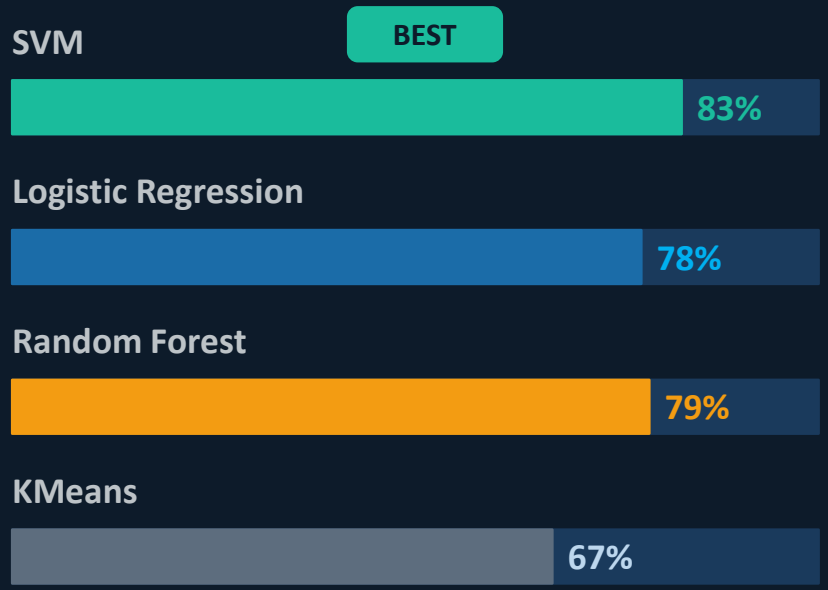
Q4: Is Glucose the most important feature?

YES

Feature importance analysis confirms: Glucose > BMI > Age in predicting diabetes

CONCLUSION

MODEL ACCURACY



KEY FINDINGS

- Glucose is the #1 predictor of diabetes onset, followed by BMI and Age
- SVM (83%) is the best classifier due to robustness against outliers in the dataset
- Patients above age 37 face a 51.49% probability of being diabetic
- Glucose > 140 raises diabetes risk to 68.75% — key clinical threshold
- Dataset imbalance (65% non-diabetic) required upsampling before model training
- No multicollinearity detected — all feature correlations below 0.70 threshold

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